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REVIEW ARTICLE

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Augmented reality for the surgeon: Systematic review

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Abstract

Introduction: Since the introduction of wearable head-up displays, there has been much interest in the surgical community adapting this technology into routine surgical practice.

Methods: We used the keywords *augmented reality* OR *wearable device* OR *head-up display* AND *surgery* using PubMed, EBSCO, IEEE and SCOPUS databases. After exclusions, 74 published articles that evaluated the utility of wearable head-up displays in surgical settings were included in our review.

Results: Across all studies, the most common use of head-up displays was in cases of live streaming from surgical microscopes, navigation, monitoring of vital signs, and display of preoperative images. The most commonly used head-up display was Google Glass. Head-up displays enhanced surgeons' operating experience; common disadvantages include limited battery life, display size and discomfort.

Conclusions: Due to ergonomic issues with dual-screen devices, augmented reality devices with the capacity to overlay images onto the surgical field will be key features of next-generation surgical head-up displays.

KEYWORDS

neuro, surgery, augmented reality, endoscopy, Google Glass, head-up display, spine, surgery, wearable device

1 | INTRODUCTION

There has been interest in using head-up displays for performing surgery since the 1990s.[1-3] While there has been some prior interest in using augmented reality and head-up displays in the operating room, currently there is no reliable commercial head-up display visualization system that is widely used among surgeons. Head-mounted displays have been reported to facilitate and expedite processes in the operating room. Starting in 2009, Liu *et al.*[4-7] found that anaesthesiologists detected changes in vital signs faster while wearing a head-mounted display during surgical procedures. Takeshita *et al.*[4] found that clinicians experienced less physical burden while performing sonographic examinations wearing a head-mounted display. It has also been shown that wearing a head-mounted display can result in reduced visual scanning; when anaesthesiologists wore head-mounted displays, they spent more time looking at the patient rather than at the anaesthesia workstation.[5] More recently, the launch of Google Glass (Google, Mountain View, CA) has resulted in great interest in adopting its use in the operating room. Its use consisted of simple video recordings of procedures with the Google Glass camera, such as capturing a first-person view of the surgeon during a surgical procedure.[6] Google Glass has also been used for displaying electronic medical records, medical imaging, checklists and vital signs, and it has been used for mobile consultation and teaching. Chimenti and Mitten[7] reported using the Google Glass display to view fluoroscopic images during percutaneous pinning of a hand fracture and

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concluded that Google Glass allowed surgeons to direct their attention toward the operative field more consistently. Another clinical study conducted by Wu *et al.*[8] demonstrated that fewer head movements were required when Google Glass was used while performing ultrasound-guided central line placement. Yoon *et al.*[9] performed 3-dimensional (3D) image-guided spine instrumentation in 10 patients with the assistance of a wearable head-up display, which resulted in less time spent placing pedicle screws as well as an overall improved operative experience for the surgeon based on post-operative surveys.

A reliable wearable device could be incorporated for use in 21 million image-guided surgeries performed across the United States each year or 118 million image-guided surgeries performed worldwide each year.[10] The widespread adoption of wearable devices for image-guided surgery can potentially lead to an improved operating experience, decreased operative time, lower rate of surgical mistakes and decreased overall costs for healthcare systems.

Wearable head-up displays for image-guided surgery could be potentially proven beneficial by saving time. It has been estimated that every minute in the operating room costs the hospital between $120 and $250.[10] If wearable devices can save time for a surgeon, then this would result in increased capacity to conduct more surgeries in operating rooms in the hospital and ultimately contribute to an increase in the hospital's net income.

Many head-up display technologies have traditionally been developed for consumer applications. However, there has been ongoing effort to develop various custom head-up display solutions for specific use in surgical cases.

In this comprehensive review, we aim to examine the usage of head-up displays in surgical operations. Although head-up display technology is generally a promising frontier in standard operating procedures, we have highlighted the various advantages and limitations of using such technology in the operating room. Evaluations with respect to efficacy as well as qualitative observations of the user experience are reviewed. In addition, we review a set of head-up display devices that are currently on the market and describe the potential advantages and disadvantages if they were to be adapted to surgical procedures.

2 | METHODS

2.1 | Search strategy

We conducted a systematic review and meta-analysis of original articles using electronic databases: PubMed, EBSCO, IEEE and SCOPUS. Literature searches were performed using *augmented reality* OR *wearable device* OR *head-up display* AND *surgery*. Only the articles written in English from 1995 through 2017 were considered.

2.2 | Selection of studies

We then followed the guidelines from Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) to identify literature to review. After 3385 publications were retrieved, 427 duplicates were identified and excluded. The following exclusion criteria were applied: (1) studies that were not original articles (reviews or commentaries), (2) studies of other wearable devices that are not head-up displays (e.g., watches or sensors), (3) studies involving the use of head-up displays for nonsurgical situations such as charting, education and data sharing (Figure 1).

The resulting 74 articles included those that evaluated the use of a head-up display in surgical procedures. We also reviewed commercially available head-up displays on the market and highlighted potential advantages and disadvantages. We included a comprehensive review of our experience of using these devices (Table 1).

3 | RESULTS

Our search resulted in 74 published articles that evaluated the utility of wearable head-up displays in the setting of the operating room. Thirty-seven of these studies involved live human surgeries while the other 37 were used in simulated operations. Wearable head-up displays were used in anaesthesia, ophthalmology, neurosurgery, urology, dermatology, spine/orthopaedic surgery, oral and maxillofacial surgery, and general surgery. Across all studies, the most common uses of head-up displays were in live streaming from endoscopy, navigation images from intraoperative magnetic resonance imaging (MRI) or computed tomography (CT), monitoring of vital signs, and display (either holographic or still images) of preoperative MRI or CT. The most commonly used head-up display in surgery is Google Glass, which is reported in 18 studies (13 with live surgeries, 5 with simulated surgeries), for various uses including video recordings and surgical navigation. Additionally, Sony stereoscopic head-mounted displays (HMZ-T2, HMZ-T1, HMI-3000MT, HMS-3000MT, HMM-3000MT; Sony, Tokyo, Japan) for live imaging were used in 11 studies in live surgeries, mainly for endoscopy.

3.1 | Current consumer head-up displays on the market

Google Glass (Google, Mountain View, CA) is a wearable display that operates on the Android mobile operating system (Figure 2A, 2B). Google Glass is equipped with a camera, accelerometer, gyroscope, GPS, Wi-Fi and Bluetooth. It is unique in that it uses a bone conduction transducer to provide audio to the wearer. It also can be controlled by voice or gesture. Google Glass has been used in various situations in surgical operations, mostly for streaming recorded videos of surgeries to remote locations.

Optinvent (Rennes, France) is a wearable display that also operates on the Android mobile operating system (Figure 2C). Optinvent (ORA-1) has two configurations – augmented reality and 'glance' mode. It is also equipped with a camera, 9-axis motion sensor, GPS, Wi-Fi and Bluetooth. It can be controlled by voice, but not gestures, unlike Google Glass.

VuFine (Sunnyvale, CA) is a wearable display that is unique in that it serves only as a display monitor (Figure 2D, 2E). VuFine is not enabled for wireless streaming; it must be connected to the image source with a high-definition multimedia interface cable. Moreover, VuFine does not have a camera or any other additional features inside.

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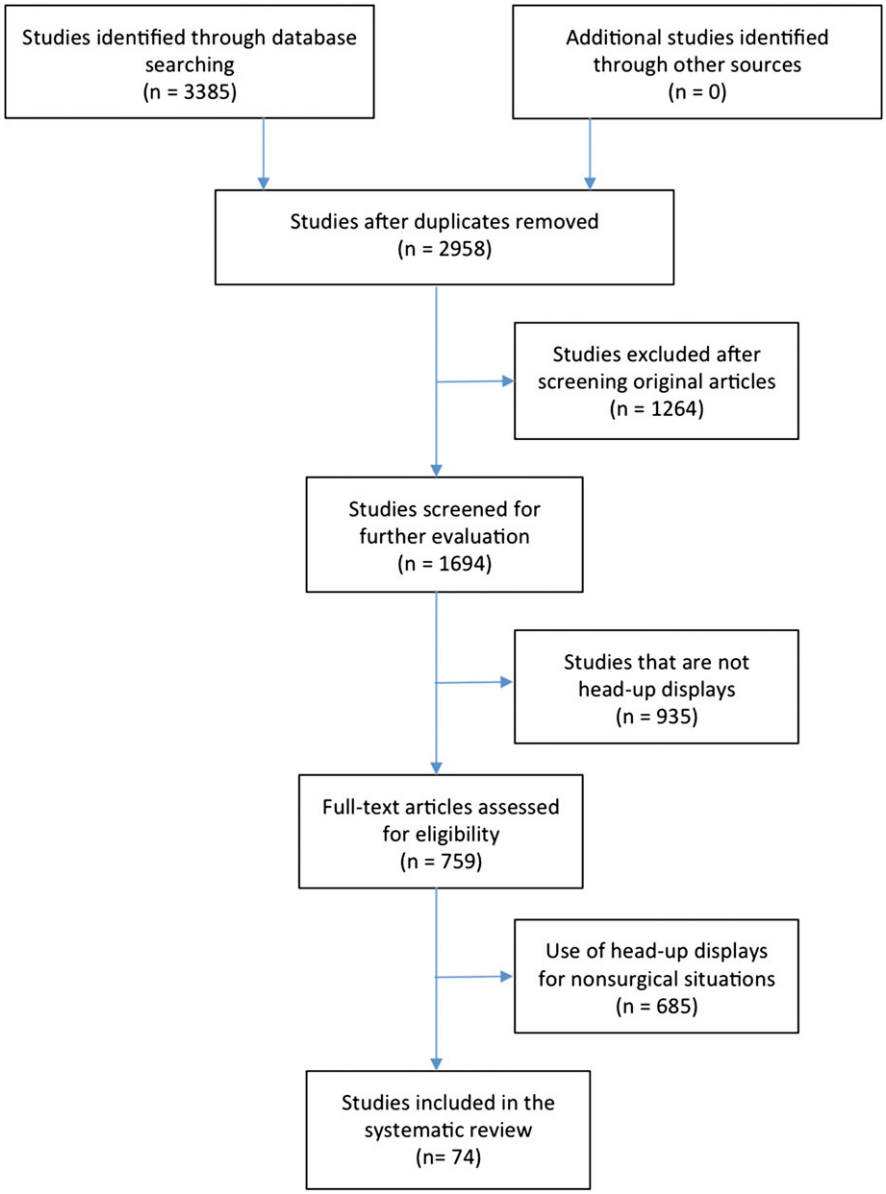

FIGURE 1 A flowchart showing inclusion and exclusion criteria using a PRISMA search strategy

Hololens (Microsoft, Redmond, WA) is a wearable head-up display that is used for augmented reality (Figure 2F). When wearing the device, the user is able to see holograms of preprogrammed computer-generated objects in the same line of vision as one's visual field. This device may potentially be useful for visualization of 3D reconstructions of preoperative planning images (e.g., from MRI or CT scans) overlaid onto the surgical field.

Oculus Rift (Oculus VR, Menlo Park, CA) is a wearable head-up display for virtual reality (Figure 2G). The display encompasses the entire visual field. The Oculus Rift has to be paired with a computer operating Microsoft Windows in order to be used. Software is written on the computer and is synced with Oculus in order for the virtual reality world to be displayed to the user. The Oculus Rift has been used for video games[11] as well as simulations for job training (e.g., for automobile technicians).[12] While Oculus Rift is a fully immersive virtual reality device, surgical loupes cannot be mounted to it.

Vuzix M300 (Vuzix, Rochester, NY) (Figure 2H) is a wearable head-up display that runs on the Android mobile operating system. The Vuzix M300 can be mounted to surgical loupes and the angles can be adjusted, which can be helpful for surgeons.

Vuzix iWear (Vuzix, Rochester, NY) is a wearable head-up display for virtual reality (Figure 2I). Like Oculus Rift, the display encompasses the entire visual field and is unable to be mounted to surgical loupes.

3.2 | Breakdown of studies by display type and surgical specialty

Figure 3 shows the distribution of studies with live humans broken down by 10 surgical specialties. The most studies were in urology (9 studies) and the fewest were in interventional radiology (1 study). Further breakdown of these studies is found below.

Figure 4 shows the distribution of studies with live humans broken down by 12 head-up display types. The most commonly used

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TABLE 1 Wearable head-up displays that are currently available in the market: strengths and weaknesses

Wearable Device (Website)	Company	Mountable to Loupes	Strengths	Weaknesses
Google Glass (google.com/glass/)	Google	Yes	Light frame and weight (35 g) Voice commands Runs Android apps	2 hour battery life Small screen size Transparent screen Maximum resolution of 640 × 360 pixels
Optinvent ORA-2 (optinvent.com/produit/ora-2)	Optinvent	No	Large screen size Resolution of 42 pixels/deg	Bulky frame Transparent screen 2-hour battery life
Vufine (vufine.com/product)	Vufine	Yes	Large screen size High resolution of 960 × 540 pixels Less transparent screen Voice commands Light weight (26 g)	Cannot be mounted on surgeon's loupes with magnet 1.5 hour battery life
Hololens (microsoft.com/microsoft-hololens/en-us)	Microsoft	No	Holographic projections onto surfaces Voice commands	Bulky frame
Oculus Rift (oculus.com)	Oculus VR	No	Large screen size Immersive user experience	Bulky frame Cannot view real world simultaneously
Vuzix M100/M300 (vuzix.com/Products/m300-smart-glasses)	Vuzix	Yes	Longer battery life (2–12 hrs) Voice commands	Bulky frame
Vuzix iWear (vuzix.com/Products/iWear-Video-Headphones	Vuzix	No	Immersive user experience Voice commands	Bulky frame Cannot view real world simultaneously

head-up display in surgery was Google Glass, which was reported in 18 studies. Of these 18 studies, 13 were in surgeries on live patients and 5 were in simulated surgeries. There are few different wearable-computing devices on the market that can be used instead of Google Glass, and there are also streaming devices, such as MedCyclops (Jacksonville, FL), that are compatible with multiple wearable head-up displays. Previously, Google Glass has been used in the operating room for recordings and live streaming of simple procedures via Google Hangout for teaching and sharing purposes.[6-8,13-22]

Figure 5 shows the distribution of studies broken down by viewing purposes for 37 live human surgeries. Live imaging of endoscopy was used the most (12 times). Head-up displays were also utilized to

FIGURE 2 Wearable head-up displays shown individually and mounted onto surgical loupes. A, B, Google Glass; C, Optinvent; D, E, Vufine; F, Microsoft Hololens; G, Oculus Rift; H, Vuzix M300; I, Vuzix iWear. Images in Figure 2A (Google Glass), Figure 2F (Microsoft Hololens) and Figure 2 G (Oculus Rift) are adopted from http://flickr.com and no changes were made to these original images. License: https://creativecommons.org/licenses/by/2.0/

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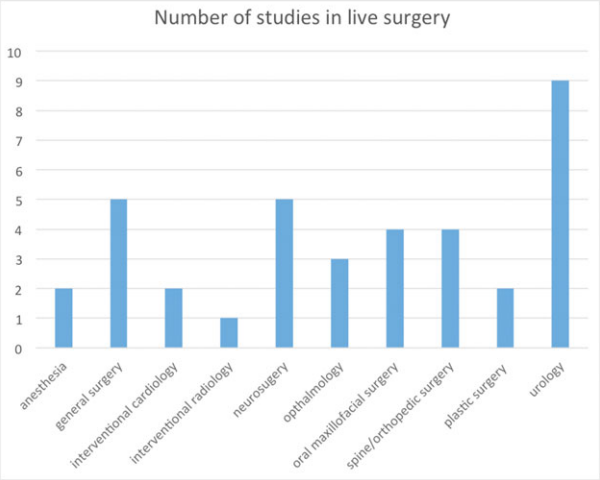

FIGURE 3 Distribution of studies by surgical specialty in 37 live human surgeries. The graph shows the distribution of studies with live humans broken down by 10 surgical specialties. The largest number of studies were performed in the field of urology

view intraoperative navigation images during surgery. These devices were also used to record operative videos or take photos during surgery.

3.3 | The use of wearable head-up displays by surgical specialty

Typically, studies evaluating the usage of head-mounted displays conduct a head-to-head comparison of operating experience using the display versus normal operation (i.e., no usage of the head-mounted display). In most cases, when the head-mounted display is used, the standard monitor is still used, so that the surgeon can default to the standard

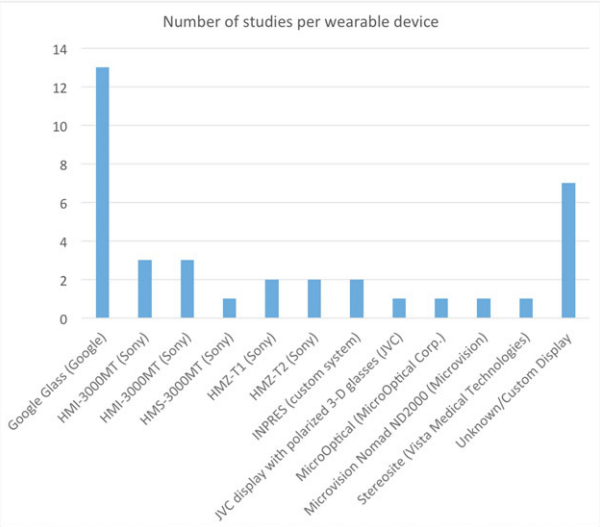

FIGURE 4 Distribution of studies by head-up display device in 37 live human surgeries. The most commonly used head-up display in surgery was Google Glass, which was reported in 13 surgical studies. Other types of wearable head-up displays were infrequently used and some authors utilized custom head-up displays that were specifically designed for their studies and were not commercially available

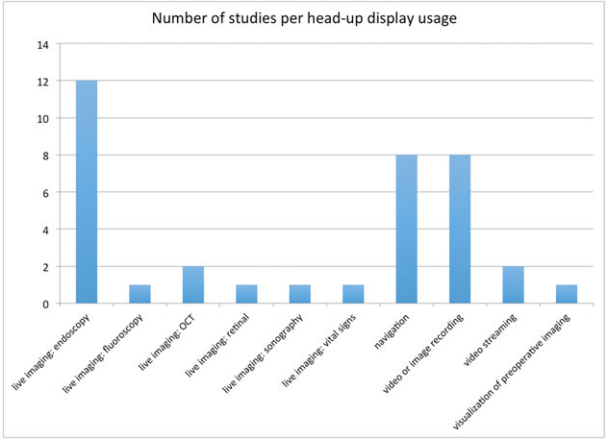

FIGURE 5 Distribution of studies by viewing purpose in 37 live human surgeries. Live imaging of endoscopy was used the most (12 times). Head-up displays were also being utilized to view navigation images. Video recording and photographing through head-up displays were also frequently used during surgery

monitor display if the head-mounted display is inadequate for carrying out the surgical procedure. User experiences are typically evaluated with pre- and postoperative surveys. Furthermore, the usefulness of the device is typically evaluated by objective measures, such as time spent or degree of attention paid (measured by number of times the standard monitor is looked at). Table 2 summarizes reported usage of wearable devices in different medical specialties, surgical application, number of studies performed and number of patients included in their studies.

3.4 | Anaesthesia

Across all the studies identified in the area of anaesthesia, 2 studies were conducted in live patients. Liu *et al.*[23] evaluated the efficacy of a head-mounted display (Nomad ND2000 HMD, Microvision, Redmond, WA) in combination with a standard anaesthesia workstation/monitor and compared it to using the standard anaesthesia monitor alone. In this study, 6 anaesthesiologists conducted procedures with and without the head-mounted display in 36 patients undergoing rigid cystoscopy, and the usage of the display resulted in more time spent looking at the surgical field instead of the standard anaesthesia workstation (55.9% vs. 51.5%; $P = 0.014$). The other study in live patients involved the Google Glass device used in tracheal intubation and laryngoscopy. Spencer *et al.*[13] report the usage of Google Glass to document perioperative airway assessment in a patient presenting to the emergency room with limited mouth opening and malocclusion due to blunt trauma.

There were 7 studies utilizing head-up displays for simulated anaesthesia procedures (6 with Nomad, 1 with Google Glass and 1 with a custom headset). Use in simulated patients included laryngeal mask airway placement[24]; monitoring of vital signs such as respiratory rate, oxygen saturation and blood pressure[14,25-28]; and phantom needle guidance in simulated peripheral nerve blocks.[29] Yoon *et al.* (unpublished) successfully performed ultrasound-guided brachial plexus nerve block through a supraclavicular approach in a patient using Google Glass (Figure 6B). The simulated studies found that usage of a head-up display was effective in shortening the time to

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TABLE 2 Studies about augmented reality in surgery, broken down by medical specialty or type of surgery and surgical application

Medical Specialty	Surgical Application	No. of Studies	No. of Patients
Anaesthesia	Rigid cystoscopy	1	36
	Tracheal intubation/laryngoscopy	1	2
Dermatology and plastic surgery	Unspecified, various	1	not specified
General surgery	Diabetic limb salvage	1	1
	Transanal microscopic tumour removal	1	1
	Endoscopic submucosal dissection	1	1
	Lung transplant	1	6
	Unspecified	1	1
Interventional cardiology	Computed tomography angiogram	2	16
Interventional radiology	Percutaneous transluminal angioplasty	1	3
Neurosurgery	Craniotomy	2	17
	Intradural lumbar biopsy	1	1
	Ventriculoperitoneal shunt	3	51
	Endoscopy	4	82
Ophthalmology	Lamellar hole, macular hole, epiretinal membrane, retinal detachment	1	48
	Vitrectromy, phacoemulsification	1	1300
	Ocular, various	1	40
Oral and maxillofacial surgery	Le Fort 1 osteotomy	1	1
	Tumour resection; osteosarcoma upper jaw	1	1
	Unknown	2	2
Spine/orthopaedic surgery	Fluoroscopy, various orthopaedic procedures	1	50
	Spine instrumentation: pedicle screw	2	11
	Spinal electromyogram	1	1
Plastic surgery	Facelift with upper and lower blepharoplasty	1	1
	Unspecified, various	1	9
Urology	Nephrectomy	4	9
	Prostatectomy	3	8
	Diagnosis and treatment: renal cyst, bladder cancer, benign prostate hyperplasia, prostate cancer	1	17
	Urothelial carcinoma treatment	1	5
	Ureteral stent	1	1
	Unspecified, various	1	31

detect adverse or risky events. Furthermore, post-procedure surveys typically showed that the head-up displays were easy to use and led to overall positive user experiences.

3.5 | Dermatology

While there are limited published accounts of head-up display usage in dermatology practice, 1 dermatologist reported using Google Glass to take photographs of selected procedures at his discretion, although specific cases were not documented.[15] Furthermore, another dermatologist suggests that Google Glass can be leveraged for lesional and histopathologic photography; perioperative photography; documentation of Mohs stages; Mohs specimen orientation; patient education regarding tumour size, defect size and location; and reconstructive surgery.[16]

3.6 | Spine/Orthopaedic surgery

In spine surgeries, there were 4 studies where a head-mounted display was used in a live surgery. In this study, the MicroOptical (MicroOptical Corporation, Westwood, MA, USA) head-up display was used for viewing intraoperative fluoroscopy during open reduction and internal fracture fixations and spinal pedicle screw placement in 50 cases.[30] The use of the display resulted in significantly fewer times that the surgeons had to turn away from the patient to view an imaging monitor. In this particular case, reducing head turns was important because surgeons may be exposed to unwanted fluoroscopy radiation by an unprotected turn of the body.

In another live surgery, head-up displays have been used for navigation as well as visualization of live endoscopic images. Yoon *et al.*[9] utilized the Google Glass head-up display to aid in pedicle screw placement in spinal instrumentation procedures in 10 patients. They found that use of the head-up display resulted in a 15% decrease in operating time (4.13 min with vs. 4.86 min without) (Figure 6A). Golab *et al.*[31] used Google Glass for monitoring of electromyogram signals during spinal surgery to assess neurologic status. Electromyogram time series signals were recorded, streamed to the Google Glass via Bluetooth and played back on the Google Glass screen in real time.[31] Lee *et al.*[17] used Google Glass and Panasonic HX-A100 as aids in spine surgery. They noted that live streaming of video to head-up displays may be limited by suboptimal wireless technology, as well as risks such as network failure, which may lead to frames being dropped or video playback being delayed. It may be important to address these issues if streaming were to be conducted in real time during surgery. Google Glass was seen as favourable due to its lightweight nature.[17]

There were 3 studies on simulated orthopaedic surgeries, for 3D visualization of landmarks in sacroiliac joint screw placement,[32,33] for fluoroscopy during fracture repair (with Google Glass), and for repair

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FIGURE 6 Examples of utilization of Google Glass head-up displays in neurosurgery and ultrasound-guided procedures. Image transfers were completed using hardware and software from MedCyclops (Jacksonville, FL). A, The surgeon is able to track the location of a pedicle screw in the right upper corner of his vision through the Google Glass screen without needing to turn his head to see the navigation screen located behind his back. B, An anaesthesiologist is able to view ultrasound images in real time without delays and complete brachial plexus nerve block without taking his eyes off the needle and the patient. He does not have to turn his head to see the ultrasound monitor that is located away from the patient. C, Wearable device can be useful in craniotomy and tumour removal. A neurosurgeon is able to view the navigation screen without taking his eyes off the patient

of various orthopaedic traumas.[34] For many orthopaedic surgeries, a head-up display system that allows both improved navigation via visualization of intraoperative MRI and 3D anatomic reconstructions of preoperative images and visualization of fluoroscopy would likely be beneficial.

3.7 | Ophthalmology

In ophthalmology, there were 3 studies that used a head-up display in live human surgery, 2 of which involved optical coherence tomography and 1 of which involved vitreoretinal imaging with 3D glasses. Eckardt and Paulo[35] evaluated the usage of a head-up display for vitreoretinal imaging with 3D glasses (46″ high-definition LCD display (JVC GD463D10UQ; Yokohama, Japan) with polarized 3D glasses) as an aid for 500 microscopic-guided phacoemulsifications and 400 image-guided vitrectomies over a span of 7 months. The intraoperative imaging camera was broadcast onto a head-up display worn by the surgeon, and this display was used to visualize the cataract or retina during the respective surgery. One of the major drawbacks witnessed in this study was insufficient image illumination, which was in part due to the 3D glasses normally worn (for the purpose of visualizing the retina) while performing the procedures. None of the procedures performed resulted in any unexpected problems or complications. Furthermore, the head-up display was shown to be effective because in all surgical cases the surgeon was not required to switch views between the head-up method and the traditional viewing method.

Carrasco-Zevallos *et al.*[36] demonstrated that usage of a stereoscopic display for 4-dimensional visualization in optical coherence tomography led to improved accuracy of simulated corneal suture placement, as measured by the difference between the depth of suture placement and the target depth (12.31% with the display compared to 43.63% without the display).

Shen *et al.*[37] implemented a custom-built stereoscopic head-up display for optical coherence tomography, and used it *in vivo* in 40 human vitreoretinal surgeries (macular surgery for epiretinal membrane, macular hole, lamellar hole and/or retinal detachment). Their device resulted in 50.0% (SD = 37.3%) higher success rate and 35.0% (SD = 24.8%) less operating time when using the stereoscopic display compared to a standard monoscopic display.[37]

In simulated studies, a head-up display was used for a mock deep anterior lamellar keratoplasty procedure,[38] as well as a surgical skill practice task where volunteers were required to use the device to aid in placing sequins onto needles on a biconvex polystyrene disk.[35]

3.8 | Neurosurgery

In the area of neurosurgery, there were 5 studies using a head-up display for surgery in live settings and 5 studies using it for simulated surgeries.

Van Lindert *et al.*[39] used a head-mounted display (Stereosite™ Visualization & Information System, Vista Medical Technologies, Inc., Carlsbad, CA) to aid in endoscopy in 269 procedures, most of which were pituitary adenoma resection (n = 81). This display covered both the left and right sides of the surgeon's field of vision but was placed several centimetres away from the eyes such that the surgeon could view the surgical field concurrently. The task was performed successfully in 268 of the procedures without the surgeon having to abandon

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the head-mounted display and use the normal imaging screen. All physicians performing the task reported reduced visual strain when using the device. A drawback mentioned in this study was the potential difficulty of hand–eye coordination for assistants when assisting the surgeon.[39]

Levy *et al.*[40] evaluated the usage of a custom-made head-up display in endoscopic imaging for 16 craniotomies, 1 intradural lumbar biopsy and 42 ventriculoperitoneal shunt placements, and followed patients for 1 year after surgery. In their study, a 280 × 650 pixel head-up display screen was used. Out of all surgeries, only 1 complication occurred, which was a transient loss of auditory evoked potentials after the endoscope came into contact with the auditory nerve.[40] Yoon *et al.*[41] utilized a novel image capture and steaming device (MedCyclops, Jacksonville, FL) to transfer navigation images from an AxiEM station (Medtronic, Minneapolis, MN) to Google Glass to aid in the placement of ventriculoperitoneal shunt in 2 patients. Diaz *et al.*[42] utilized the same image capture and streaming device (MedCyclops, Jacksonville, FL) to perform craniotomy and resection of supratentorial meningioma in 1 patient (Figure 6C). In both studies, the wearable device allowed for continuous monitoring of neuronavigation images in the right upper corner of the surgeon's visual field without the need for the surgeon to turn his head to view the navigation monitors. Additionally, Levy *et al.*[43] adopted a stereoscopic head-mounted display for aiding in microsurgical and endoscopic procedures. They found that the use of the display did not lengthen (or significantly shorten) the surgery, and that it remained comfortable to wear for a period of 1 to 4 hours.

Regarding simulated surgical procedures in neurosurgery, head-up displays have been used for visualizing endoscope images[44] and stereoscopic images from other imaging cameras,[45] locating steel spheres with a bayonet probe during simulated craniotomies,[46] visualizing preoperative MRI and CT images,[47] and visualizing vertebral tissue in simulated pedicle screw placement.[48]

3.9 | Urology

Nine studies utilized head-up displays for urologic procedures. In a majority of the cases they were used for the purpose of endoscope-guided surgeries. The most commonly used head-up display devices were various versions of Sony's head-mounted 3D viewer (HMI-3000MT, HMZ-T2, HMZ-1 and HMS-3000MT (Sony Corporation, Tokyo, Japan)), which captures imaging video normally picked up from endoscopes (e.g., Zeiss surgical endoscope). Eight of the human studies used the Sony head-mounted display devices, while 1 study used the Google Glass.

The majority of studies conducted in humans involved using the head-mounted display to visualize endoscopic images in real time. In these cases, the head-mounted display was connected by wire to surgical endoscopes, and any video captured by the endoscope camera was sent directly to the head-mounted display. Procedures performed included radical nephrectomy,[49,50] partial gasless nephrectomy,[51,52] nephroureterectomy,[53] urethral stents,[54] cyst removal, and diagnosis and treatment of neoplasms, such as benign prostate hyperplasia,[53,55,56] prostatic carcinoma and urothelial carcinoma.[55] Overall, the studies typically concluded that head-up displays make the surgeries more efficient and easier to conduct. Some endoscopic examinations, such as those for partial cystectomy, may take up to 5 hours to complete,[55] so it is important that ergonomic concerns are taken into account before adopting the use of head-up displays.

One study involved usage of sonography for image-guided diagnosis and treatment of renal cysts, bladder cancer, benign prostatic hyperplasia and prostate cancer in 17 different cases handled by 5 urologic surgeons. While the study found that HMZ-T2 devices had greater diagnostic performance compared to the normal method ($P < 0.0001$), user experience surveys revealed some negative aspects about the experience, such as pain caused in wearing the devices and unexpected shifting in focus due to placement of the device on the head.[4] Furthermore, this study was conducted in sonographic examinations of the genitourinary tract, which normally take about 5 minutes. It is difficult to evaluate the utility of these head-mounted displays in longer surgical operations since the study was conducted in such short intervals.

3.10 | Surgical oncology

In the field of surgical oncology, there were no studies done using a head-up display for surgery in humans. However, there were 3 studies in animal tissues (*in vivo* and *ex vivo*). Mondal *et al.*[57] developed a binocular goggle head-up display system to visualize fluorescence imaging results to identify sentinel lymph nodes in mastectomy procedures in a study on 15 animal models. Shao *et al.*[58] used a Google Glass device to visualize *ex vivo* tissue in a simulated situation for cancer resection surgery. Zhang *et al.*[59] developed a wearable Google Glass-based system (Google Glass attached to surgical goggles) for visualizing lesions using tissue-stimulating phantoms and an *ex vivo* model of human breast tissue.

3.11 | Plastic surgery

There are 2 published studies of head-up display usage in plastic surgery on live patients in which Google Glass was used. In 1 study, 9 plastic surgeons wore the Google Glass display for capturing photographs and videos while performing surgical procedures. In a post-procedure survey where questions were scored on a scale from 1 to 5, the surgeons rated the comfort of the device at 4.56 and satisfaction of usage at 3.78.[18] In the other study, Google Glass was used to record pictures and videos while a surgeon was performing a blepharoplasty.[19] Some drawbacks of Google Glass that have been brought up by plastic surgeons include the inability to vertically adjust the camera, as well as inadequate illumination compared to traditionally used illumination sources for visualization of wounds in surgery.[20]

3.12 | Otolaryngology

There were no studies conducted in live human patients regarding otolaryngology procedures with head-up displays. One study evaluated the usage of an augmented reality surgical navigation system by 50 otolaryngologists who performed an endoscopic surgical navigation exercise on a cadaver model. The image of a CT contour overlaid onto the endoscopic image was displayed on a head-up display during the procedure and the otolaryngologists were instructed to identify

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tumours. In this study, it was found that the usage of a combination of the augmented reality device and a submonitor for visualization was more effective than just an augmented reality device alone.[60] Since head-up displays have been shown to be effective in other types of surgery involving endoscopes (e.g., urology and neurosurgery), future research should aim to evaluate and adopt head-up displays for otolaryngology.

3.13 | Oral and maxillofacial surgery

In oral and maxillofacial surgery, there were 4 studies involving usage of a head-up display for preoperative planning and intraoperative navigation in human patients (typically via visualization of the calculated planning data) in order to reduce operative risk.

A previously proposed augmented reality system, INPRES, consists of a see-through head-mounted display allowing a visual overlay of relevant information to help guide the surgeon during an operation.[61] The INPRES augmented reality system was first used for craniofacial surgery in a study by Sudra *et al.*[62] Challenges associated with utilizing such systems included difficulty with registration and calibration of the virtual data with the real-world surgical field. A similar system was used in performing a Le Fort I osteotomy procedure, in which a visual overlay of the preplanned osteotomy line was superimposed on the surgeon's view of the visual field.[63] In regard to tumour resection, Wagner *et al.*[64] proposed a wearable head-up display solution, which displays preoperative coordinate registration images for the purpose of aiding in identifying the tumour target more accurately. Newer devices such as Google Glass have not been used in any cases of oral and maxillofacial surgery.

3.14 | Cardiothoracic surgery

There was 1 study in simulated cardiothoracic surgery. Austin *et al.* evaluated the use of a custom head-up display and software as an aid in assimilation and processing of information for cardiopulmonary bypass surgery to visualize multiple physiologic parameters on the screen in a hexagon-shaped visualization. Physiologic signals visualized included cardiac index, peripheral vascular resistance, haematocrit, dynamic operating blood level, venous saturation and mean arterial pressure.[27]

3.15 | Interventional cardiology

The 2 studies in interventional cardiology were both performed on live patients. The first case of CT-guided percutaneous revascularization using a wearable head-up display was demonstrated by Opolski *et al.*[65] In this study, Google Glass was used to visualize coronary CT angiography reconstructions (computed by the CT scanner and software and projected onto Google Glass) of the right coronary artery during the procedure in 1 patient.[65] In a separate study to test image quality, Duong *et al.*[66] used the Google Glass to record 15 angiography procedures, where the Google Glass camera was used to take videos of the angiogram monitor. The authors found no significant difference in picture quality between the video recorded by Google Glass and the images of the original angiograms, thus suggesting that Google Glass may be useful in cardiac catheterization.[66]

3.16 | Interventional radiology

One study in the area of interventional radiology was identified. Vorraber *et al.*[67] demonstrated the use of Google Glass to visualize vital signs in three patients undergoing percutaneous transluminal angioplasty. Post-procedure feedback showed that Google Glass was helpful for raising situational awareness in the surgeons and that there were no negative influences of the Google Glass device that interfered with the procedure.[67]

3.17 | Vascular surgery

No studies in live human subjects were conducted with head-up displays in the subfield of vascular surgery. There was 1 study involving a simulated procedure, in which Gao *et al.*[68] developed an augmented-reality solution for vascular catheterization in the leg, in which preoperative guidance images of the vascularization and planned catheter trajectory were displayed on a head-up display while the surgeon kept his attention on the surgical field. The device was tested by 2 surgeons on a phantom model, and both surgeons found that the path displayed on the head-up display was useful for conducting the procedure.[68]

3.18 | General surgery

In the area of general surgery, there were 5 studies done in live human patients and 9 studies done in simulated surgeries.

Among the studies done in live human patients, 2 studies were for the purpose of live imaging with endoscopy and 2 studies used the Google Glass device to record and stream videos of the surgery.

Morisawa *et al.*[69] conducted the only study of head-up display usage for gastrointestinal surgery in live patients using the Sony HMM 300MT (Sony, Tokyo, Japan) head-mounted display for endoscopic submucosal dissection in 6 patients and compared this to using a normal display for the operation in 104 patients. While the surgeons wearing the device reported a positive operating experience, the use of the device did not result in significant differences in procedure time, operation accuracy (100% resection in all cases) or complication rates.[69] Koesveld *et al.*[70] evaluated the use of a head-mounted display in transanal endoscopic microsurgery, in which a resection of a villous adenoma was successfully performed.

For the purposes of video streaming, Armstrong *et al.*[21] demonstrated use of the Google Glass device as an aid in performing diabetic limb salvage surgery. In this study, the surgeons wore Google Glass when treating a plantar wound caused by diabetic foot ulcer. Google Glass was then used to record a video when the surgeon performed the operation, which simultaneously was wirelessly streamed to a mobile phone device and remote computer via internet connection.[21] Baldwin *et al.*[71] used the Google Glass device to record video and still shots from the surgeon's point of view in 4 transplant-related operations: a right single lung recovery, a left single lung recovery, and 2 bilateral lung harvests. The authors reported that the ability to record a video and stream it in real time is important for transplant surgery because oftentimes the donor and recipient bodies need to be operated on simultaneously. Therefore, this feature of Google

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Glass allows for swift communication between multiple members of the transplant team.[71]

There were 9 published studies on simulated procedures in the area of general surgery, for purposes such as live imaging via ultrasound and endoscopy,[8,72-75] intraoperative guidance for needle placement,[76] navigation for general surgeries,[77] intubation,[78] and radio-frequency ablation.[79]

4 | DISCUSSION

4.1 | Unmet clinical needs addressable by head-up displays

4.1.1 | Attention and focus

Currently, surgeons must shift their concentration between the patient and the visual display, which is usually located on a wall placed away from the patient. Doing so may lead to unintended movements when performing image-guided surgeries, which can lead to unfavourable outcomes depending on the nature of the surgery. For example, a surgeon performing a brain tumour resection may need to manoeuvre a probe to the appropriate part of the brain. However, since the surgeon relies on the image shown on the medical imaging monitor, he is more likely to navigate the probe on an unintended trajectory during the split seconds when he takes his attention off the surgical field. Multiple iterations of mistakes can lead to accumulated disruption in tissue, which may be harmful to the patient.

4.1.2 | Procedure planning

At the current state of technology, surgical planning is performed based on 2-dimensional images of preoperative imaging modalities such as MRI and CT. A surgeon must mentally reconstruct 3D images, then plan and execute the location and extent of surgical incision, as well as extent of tumour resection. For example, during craniotomy, the surgeon maps out the skin incision based on known anatomic landmarks and 2-dimensional preoperative imaging studies. Augmented reality devices capable of holographic projections may be highly useful in assisting such surgeries. One of the devices furthest along in development is the Hololens (Microsoft, Redmond, WA), which projects a hologram in a user's line of vision when the head-mounted device is worn. One example of a useful teaching tool not involving head-mounted displays or holograms is the Anatomage Virtual Dissection Table with Invivo5 software (Anatomage, San Jose, CA),* which displays holographic projections of human organs onto a touchscreen table. This device may be useful for teaching purposes as well as preoperative mapping and planning.

4.1.3 | Time savings

One of the benefits of wearable devices for image-guided surgery lies in time saving, considering that an estimated cost for one minute in the operating room is between $120 and $250.[10] If wearable devices can save time for a surgeon, one possible result could be an increased capacity for taking on more surgeries in operating rooms in the hospital, ultimately contributing to an increase in the hospital's net income. The use of a wearable head-up display led to a 15% decrease in operating time for image-guided spine instrumentation in one study.[9] Larger and more robust trials are needed to further define the time saving afforded by utilization of wearable head-up display systems in live surgeries.

4.1.4 | Benefits of wearable head-up displays

Most studies we reviewed have found that wearable head-up displays for surgery usually result in improved ergonomics compared to traditional procedures in which the surgeon is required to turn his or her head repeatedly to shift visual focus between the surgical field and the imaging monitor.[35] Head-up displays allow surgeons to concentrate on the task at hand without having to turn their heads away from the surgical field constantly.[5,6]

Some head-up displays, such as Google Glass, have a camera and Wi-Fi that allow surgeons to record procedures and stream them out to other parties. The recorded videos can aid in charting and documentation purposes, and a Wi-Fi streaming capability can facilitate communication between parties. For example, Google Glass has been used to stream videos of surgeries from the operating room to a remote party.[21] They may also be useful for intraoperative consultations as well as teaching and assistance for surgery residents and other personnel.[71]

In the field of anaesthesia and intensive care units, fast streams of information are crucial in order to increase situational awareness of abnormal events. Often patient monitoring systems are equipped with alarms; however, a head-up display would serve as a catalyst to alert physicians in the event of emergency to shorten the time between the alarm and the physician's awareness of the problem.[13,23]

Additionally, studies quantifying surgeons' experiences using wearable head-up displays demonstrated overall high satisfaction. For example, in Yoon *et al.*'s study, surgeons' experience with a head-up display during 10 spine surgeries showed an increasing satisfaction as the surgeons became more familiar with the device. Positive experience was recorded in 79% of the post-survey questions.[9] Moreover, in Eckardt's study using a head-up display for vitreoretinal procedures, 91.7% of the medical professionals preferred the head-up display to the traditional method.[35] Even though a majority of literature reviewed in this paper mentioned physician satisfaction using the device, many did not quantify the experience. Future studies should use appropriate measures to quantify clinical experience in order to better compare the efficacy of new devices to standard methods.

4.1.5 | Drawbacks of wearable head-up displays

The size of the screen is cited as one of the drawbacks. One of the major problems that arise in many studies involves eye strain (asthenopia) associated with concentrating on a small screen for prolonged periods of time.[35]

As with viewing of any sort of imaging monitor, prolonged usage can lead to visual discomfort. This is a problem with not only wearable devices but also imaging devices in general, including medical imaging monitors that are traditionally used.[40,80]

*http://www.anatomage.com/portfolio-item/anatomage-table/

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Battery power is another drawback of any head-up display that must be addressed because of long operative times inherent to certain surgeries. For example, when used for streaming or broadcasting, the Google Glass battery life can be as little as 30–40 minutes.[6,22]

While augmented reality brings about many benefits, such as improved situational awareness in surgical procedures, augmented reality involving head-up displays may draw too much focus away from the operating field.[6] This drawback may also be more applicable to certain types of surgery than others. In addition, there is a concern that since head-up displays such as Google Glass were developed for the general public, their voice recognition system might not recognize medical terms. In surgical use, most people did not utilize the voice recognition capabilities of their head-up displays, but this limitation should be addressed if voice recognition were to be used during surgery (e.g., for annotation, hands-free commands, etc.).

Finally, privacy is an important aspect of wearable head-up displays that must be addressed before they are adopted freely in the clinic.[6] To comply with US regulations, information must be transmitted in Health Insurance Portability and Accountability Act-compliant channels, preferably encrypted over the network.

4.1.6 | Areas of improvement for head-up displays

An area of improvement that has persistently been an issue with head-up display is inadequate brightness. In surgical applications, this is especially important because items such as tumours in the brain or polyps in the colon are best visualized in a bright image.

In addition, current wearable head-up display screens require surgeons to divert their visual attention from the primary task to the images on the head-up display. Even though the head-up displays bring critical images closer so that the surgeons do not have to physically move their heads, this dual screen requires switching attention from one screen to the surgical task. The next-generation wearable device should be able to holographically project images onto the surgical field to avoid this problem. For instance, if the location of a brain tumour can be projected onto the surgical field, the surgeon does not have to take his attention away to view navigation images. These holographic navigation images must be accurately projected and correlated with anatomic landmarks, which requires development of innovative algorithms and new surgical tools. Recently, a video projector (PicoPix 1020, Philips, Amsterdam, Netherlands) was used to project neuronavigation images directly onto the head, skull and brain surfaces with a mean error of 0.8 ± 0.25 mm when compared with the standard navigation system.[81] Translating this technology from a projector to a wearable head-up display would be the next step.

Lastly, it may be important to understand patient outcomes in surgeries involving head-up displays. Such improvements in outcomes would give healthcare providers confidence in adopting these displays for their surgeries. Currently, none of the studies on live human surgeries we reviewed had conducted an outcome analysis, such as a Kaplan–Meier survival analysis. Such statistical study in conjunction with quantifiable measures of cost-effectiveness, practicality and advantages should be addressed in future studies.

5 | CONCLUSION

There are a variety of unmet clinical needs in image-guided surgery, including attention and focus issues with the surgeon, ease of displaying relevant information for purposes of raising situational awareness, and time savings. These unmet clinical needs can be addressed with wearable head-mounted devices, which are just beginning to be explored in surgery. Future efforts should design appropriate studies with specific endpoint criteria and statistical analysis of quantifiable outcome measures to rigorously evaluate the cost-effectiveness, safety and benefit of these devices in a variety of types of surgery and settings.

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